

DoSk

July 3, 2025

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[12]: import numpy as np
import matplotlib.pyplot as plt
E_F = 12.2400 # Fermi energy in eV
def load_data(filename, up_index=2, down_index=3):

    # load data
    data = np.loadtxt(filename)

    E = np.unique(data[:, 1])
    k = np.unique(data[:, 0])

    dosup = np.zeros([len(k), len(E)])
    dosdown = np.zeros([len(k), len(E)])

    for i in range(len(data)):
        E_index = int(i % len(E))
        k_index = int(data[i][0] - 1)

        dosup[k_index, E_index] = data[i][up_index]
        dosdown[k_index, E_index] = data[i][down_index]
        if dosup[k_index, E_index] < 0: dosup[k_index, E_index] = 0
        if dosdown[k_index, E_index] < 0: dosdown[k_index, E_index] = 0

    return k, E, dosup, dosdown

[13]: k, E, DoStotalup, DoStotaldown = load_data('Ru_pdos.k.dat.pdos_tot')

k, E, DoSRu1up, DoSRu1down = load_data('Ru_pdos.k.dat.pdos_atm#1(Ru1)_wfc#4(d)')
k, E, DoSRu2up, DoSRu2down = load_data('Ru_pdos.k.dat.pdos_atm#2(Ru2)_wfc#4(d)')

DoSRuup = DoSRu1up + DoSRu2up
DoSRudown = DoSRu1down + DoSRu2down

k, E, DoS01up, DoS01down = load_data('Ru_pdos.k.dat.pdos_atm#3(0)_wfc#2(p)')
k, E, DoS02up, DoS02down = load_data('Ru_pdos.k.dat.pdos_atm#4(0)_wfc#2(p)')
k, E, DoS03up, DoS03down = load_data('Ru_pdos.k.dat.pdos_atm#5(0)_wfc#2(p)')
k, E, DoS04up, DoS04down = load_data('Ru_pdos.k.dat.pdos_atm#6(0)_wfc#2(p)')
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DoS0up = DoS01up + DoS02up + DoS03up + DoS04up
DoS0down = DoS01down + DoS02down + DoS03down + DoS04down

k, E, DoSRu1dz2up, DoSRu1dz2down = load_data('Ru_pdos.k.dat.
↳pdos_atm#1(Ru1)_wfc#4(d)', up_index=4, down_index=5)
k, E, DoSRu2dz2up, DoSRu2dz2down = load_data('Ru_pdos.k.dat.
↳pdos_atm#2(Ru2)_wfc#4(d)', up_index=4, down_index=5)
DoSRudz2up = DoSRu1dz2up + DoSRu2dz2up
DoSRudz2down = DoSRu1dz2down + DoSRu2dz2down

k, E, DoSRu1dxxup, DoSRu1dxxdown = load_data('Ru_pdos.k.dat.
↳pdos_atm#1(Ru1)_wfc#4(d)', up_index=6, down_index=7)
k, E, DoSRu2dxxup, DoSRu2dxxdown = load_data('Ru_pdos.k.dat.
↳pdos_atm#2(Ru2)_wfc#4(d)', up_index=6, down_index=7)
DoSRudxxup = DoSRu1dxxup + DoSRu2dxxup
DoSRudxxdown = DoSRu1dxxdown + DoSRu2dxxdown

k, E, DoSRu1dzyup, DoSRu1dzydown = load_data('Ru_pdos.k.dat.
↳pdos_atm#1(Ru1)_wfc#4(d)', up_index=8, down_index=9)
k, E, DoSRu2dzyup, DoSRu2dzydown = load_data('Ru_pdos.k.dat.
↳pdos_atm#2(Ru2)_wfc#4(d)', up_index=8, down_index=9)
DoSRudzyup = DoSRu1dzyup + DoSRu2dzyup
DoSRudzydown = DoSRu1dzydown + DoSRu2dzydown

k, E, DoSRu1dx2y2up, DoSRu1dx2y2down = load_data('Ru_pdos.k.dat.
↳pdos_atm#1(Ru1)_wfc#4(d)', up_index=10, down_index=11)
k, E, DoSRu2dx2y2up, DoSRu2dx2y2down = load_data('Ru_pdos.k.dat.
↳pdos_atm#2(Ru2)_wfc#4(d)', up_index=10, down_index=11)
DoSRudx2y2up = DoSRu1dx2y2up + DoSRu2dx2y2up
DoSRudx2y2down = DoSRu1dx2y2down + DoSRu2dx2y2down

k, E, DoSRu1dxyup, DoSRu1dxydown = load_data('Ru_pdos.k.dat.
↳pdos_atm#1(Ru1)_wfc#4(d)', up_index=12, down_index=13)
k, E, DoSRu2dxyup, DoSRu2dxydown = load_data('Ru_pdos.k.dat.
↳pdos_atm#2(Ru2)_wfc#4(d)', up_index=12, down_index=13)
DoSRudxyup = DoSRu1dxyup + DoSRu2dxyup
DoSRudxydown = DoSRu1dxydown + DoSRu2dxydown

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[14]: xtik = np.array([0.0000, 0.5000, 1.2248, 1.7248, 2.4496, 3.1567, 3.8814, 4.
↳5886]) * len(k) / 4.5886
xtiklabel = np.array(['$\Gamma$', 'X', 'R', 'Z', '$\Gamma$', 'M', 'A', 'Z'])

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<>:2: SyntaxWarning: invalid escape sequence '\G'
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C:\Users\ASUS\AppData\Local\Temp\ipykernel_13680\905856440.py:2: SyntaxWarning:
invalid escape sequence '\G'
    xtiklabel = np.array(['$\Gamma$', 'X', 'R', 'Z', '$\Gamma$', 'M', 'A', 'Z'])
C:\Users\ASUS\AppData\Local\Temp\ipykernel_13680\905856440.py:2: SyntaxWarning:
invalid escape sequence '\G'
    xtiklabel = np.array(['$\Gamma$', 'X', 'R', 'Z', '$\Gamma$', 'M', 'A', 'Z'])

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[17]: cmr = plt.cm.magma_r
cmr.set_bad(color='white')
fig, ax = plt.subplots(1,2,figsize=(15, 6),sharey=True, tight_layout=True)

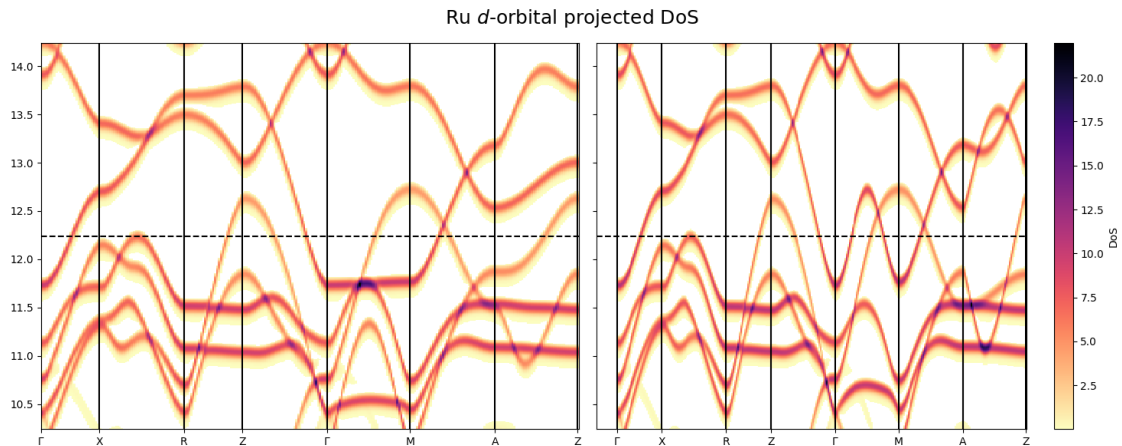
band_mask = DoSRuup < 0.01
DoSRuup[band_mask] = np.nan # Mask values below threshold
ax[0].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[0].set_ylim(E_F - 2, E_F + 2)
ax[0].pcolormesh(k, E, DoSRuup.T, cmap=cmr)
ax[0].set_xticks(xtik)
ax[0].set_xticklabels(xtiklabel)
for i in range(len(xtik)):
    ax[0].axvline(x = xtik[i], color='black', linestyle='-')
    ax[1].axvline(x = xtik[i], color='black', linestyle='-')
band_mask = DoSRudown < 0.01
DoSRudown[band_mask] = np.nan # Mask values below threshold
ax[1].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[1].pcolormesh(k, E, DoSRudown.T, cmap=cmr)
ax[1].set_xticks(xtik)
ax[1].set_xticklabels(xtiklabel)
plt.colorbar(ax[1].collections[0], ax=ax[1], label='DoS')
fig.suptitle('Ru d-orbital projected DoS', fontsize=18)

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[17]: Text(0.5, 0.98, 'Ru d-orbital projected DoS')

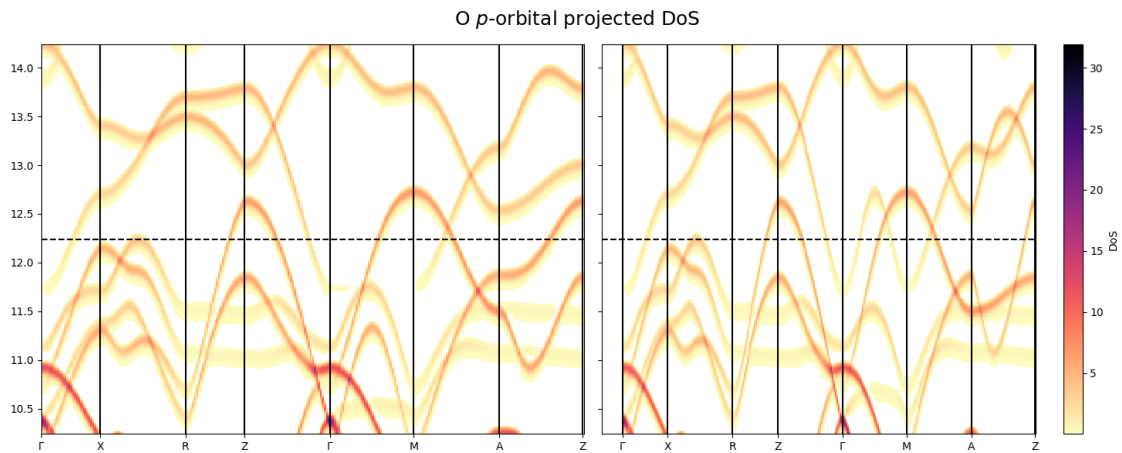
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[21]: cmr = plt.cm.magma_r
cmr.set_bad(color='white')
fig, ax = plt.subplots(1,2,figsize=(15, 6),sharey=True, tight_layout=True)

band_mask = DoSUp < 0.01
DoSUp[band_mask] = np.nan # Mask values below threshold
ax[0].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[0].set_ylim(E_F - 2, E_F + 2)
ax[0].pcolormesh(k, E, DoSUp.T, cmap=cmr)
ax[0].set_xticks(xtik)
ax[0].set_xticklabels(xtiklabel)
for i in range(len(xtik)):
    ax[0].axvline(x = xtik[i], color='black', linestyle='-')
    ax[1].axvline(x = xtik[i], color='black', linestyle='-')
band_mask = DoSDown < 0.01
DoSDown[band_mask] = np.nan # Mask values below threshold
ax[1].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[1].pcolormesh(k, E, DoSDown.T, cmap=cmr)
ax[1].set_xticks(xtik)
ax[1].set_xticklabels(xtiklabel)
plt.colorbar(ax[1].collections[0], ax=ax[1], label='DoS')
fig.suptitle('O $p$-orbital projected DoS', fontsize=18)
```

[21]: Text(0.5, 0.98, 'O \$p\$-orbital projected DoS')



```
[22]: cmr = plt.cm.magma_r
cmr.set_bad(color='white')
fig, ax = plt.subplots(1,2,figsize=(15, 6),sharey=True, tight_layout=True)
```

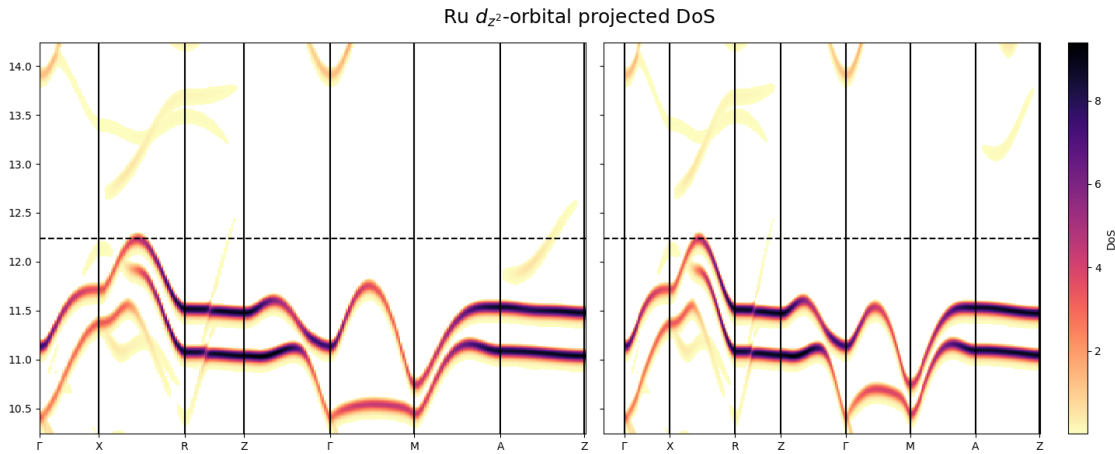
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band_mask = DoSRudz2up < 0.01
DoSRudz2up[band_mask] = np.nan # Mask values below threshold
ax[0].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[0].set_ylim(E_F - 2, E_F + 2)
ax[0].pcolormesh(k, E, DoSRudz2up.T, cmap=cmr)
ax[0].set_xticks(xtik)
ax[0].set_xticklabels(xtiklabel)
for i in range(len(xtik)):
    ax[0].axvline(x = xtik[i], color='black', linestyle='-')
    ax[1].axvline(x = xtik[i], color='black', linestyle='-')

band_mask = DoSRudz2down < 0.01
DoSRudz2down[band_mask] = np.nan # Mask values below threshold
ax[1].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[1].pcolormesh(k, E, DoSRudz2down.T, cmap=cmr)
ax[1].set_xticks(xtik)
ax[1].set_xticklabels(xtiklabel)
plt.colorbar(ax[1].collections[0], ax=ax[1], label='DoS')
fig.suptitle('Ru  $d_{z^2}$ -orbital projected DoS', fontsize=18)

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[22]: `Text(0.5, 0.98, 'Ru  $d_{z^2}$ -orbital projected DoS')`



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[23]: cmr = plt.cm.magma_r
cmr.set_bad(color='white')
fig, ax = plt.subplots(1,2,figsize=(15, 6),sharey=True, tight_layout=True)

band_mask = DoSRudzxup < 0.01
DoSRudzxup[band_mask] = np.nan # Mask values below threshold
ax[0].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')

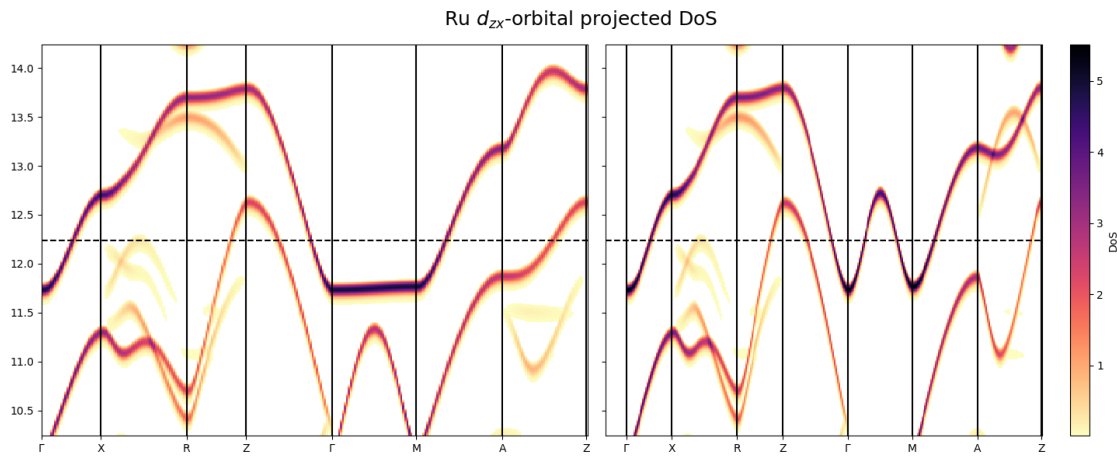
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ax[0].set_ylim(E_F - 2, E_F + 2)
ax[0].pcolormesh(k, E, DoSRudzxup.T, cmap=cmr)
ax[0].set_xticks(xtik)
ax[0].set_xticklabels(xtiklabel)
for i in range(len(xtik)):
    ax[0].axvline(x = xtik[i], color='black', linestyle='-')
    ax[1].axvline(x = xtik[i], color='black', linestyle='-')
band_mask = DoSRudzxdown < 0.01
DoSRudzxdown[band_mask] = np.nan # Mask values below threshold
ax[1].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[1].pcolormesh(k, E, DoSRudzxdown.T, cmap=cmr)
ax[1].set_xticks(xtik)
ax[1].set_xticklabels(xtiklabel)
plt.colorbar(ax[1].collections[0], ax=ax[1], label='DoS')
fig.suptitle('Ru  $d_{zx}$ -orbital projected DoS', fontsize=18)

```

[23]: `Text(0.5, 0.98, 'Ru  $d_{zx}$ -orbital projected DoS')`



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[19]: cmr = plt.cm.magma_r
cmr.set_bad(color='white')
fig, ax = plt.subplots(1,2,figsize=(15, 6),sharey=True, tight_layout=True)

band_mask = DoSRudzyup < 0.01
DoSRudzyup[band_mask] = np.nan # Mask values below threshold
ax[0].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[0].set_ylim(E_F - 2, E_F + 2)
ax[0].pcolormesh(k, E, DoSRudzyup.T, cmap=cmr)
ax[0].set_xticks(xtik)
ax[0].set_xticklabels(xtiklabel)
for i in range(len(xtik)):
    ax[0].axvline(x = xtik[i], color='black', linestyle='-')

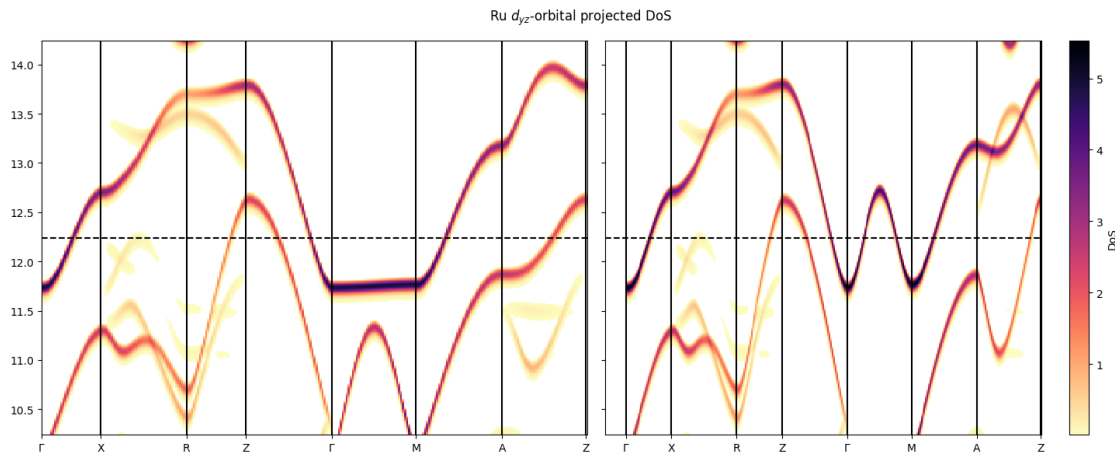
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    ax[1].axvline(x = xtik[i], color='black', linestyle='-')
band_mask = DoSRudzydown < 0.01
DoSRudzydown[band_mask] = np.nan # Mask values below threshold
ax[1].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[1].pcolormesh(k, E, DoSRudzydown.T, cmap=cmr)
ax[1].set_xticks(xtik)
ax[1].set_xticklabels(xtiklabel)
plt.colorbar(ax[1].collections[0], ax=ax[1], label='DoS')
fig.suptitle('Ru $d_{yz}$-orbital projected DoS')

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[19]: Text(0.5, 0.98, 'Ru \$d_{yz}\$-orbital projected DoS')



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[24]: cmr = plt.cm.magma_r
cmr.set_bad(color='white')
fig, ax = plt.subplots(1,2,figsize=(15, 6),sharey=True, tight_layout=True)

band_mask = DoSRudx2y2up < 0.01
DoSRudx2y2up[band_mask] = np.nan # Mask values below threshold
ax[0].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[0].set_ylim(E_F - 2, E_F + 2)
ax[0].pcolormesh(k, E, DoSRudx2y2up.T, cmap=cmr)
ax[0].set_xticks(xtik)
ax[0].set_xticklabels(xtiklabel)
for i in range(len(xtik)):
    ax[0].axvline(x = xtik[i], color='black', linestyle='-')
    ax[1].axvline(x = xtik[i], color='black', linestyle='-')
band_mask = DoSRudx2y2down < 0.01
DoSRudx2y2down[band_mask] = np.nan # Mask values below threshold
ax[1].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[1].pcolormesh(k, E, DoSRudx2y2down.T, cmap=cmr)
ax[1].set_xticks(xtik)

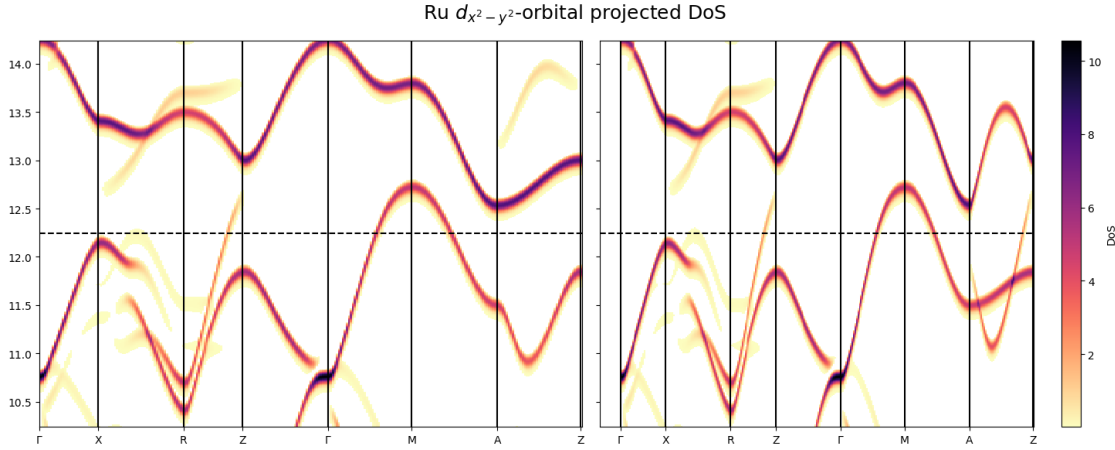
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ax[1].set_xticklabels(xtiklabel)
plt.colorbar(ax[1].collections[0], ax=ax[1], label='DoS')
fig.suptitle('Ru  $d_{x^2-y^2}$ -orbital projected DoS', fontsize=18)

```

[24]: `Text(0.5, 0.98, 'Ru  $d_{x^2-y^2}$ -orbital projected DoS')`



```

[25]: cmr = plt.cm.magma_r
cmr.set_bad(color='white')
fig, ax = plt.subplots(1,2,figsize=(15, 6),sharey=True, tight_layout=True)

band_mask = DoSRudxyup < 0.01
DoSRudxyup[band_mask] = np.nan # Mask values below threshold
ax[0].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[0].set_ylim(E_F - 2,E_F + 2)
ax[0].pcolormesh(k, E, DoSRudxyup.T, cmap=cmr)
ax[0].set_xticks(xtik)
ax[0].set_xticklabels(xtiklabel)
for i in range(len(xtik)):
    ax[0].axvline(x = xtik[i], color='black', linestyle='-')
    ax[1].axvline(x = xtik[i], color='black', linestyle='-')
band_mask = DoSRudxydown < 0.01
DoSRudxydown[band_mask] = np.nan # Mask values below threshold
ax[1].axhline(y = E_F, label='Fermi level', color='black', linestyle='--')
ax[1].pcolormesh(k, E, DoSRudxydown.T, cmap=cmr)
ax[1].set_xticks(xtik)
ax[1].set_xticklabels(xtiklabel)
plt.colorbar(ax[1].collections[0], ax=ax[1], label='DoS')
fig.suptitle('Ru  $d_{xy}$ -orbital projected DoS', fontsize=18)

```

[25]: `Text(0.5, 0.98, 'Ru  $d_{xy}$ -orbital projected DoS')`

